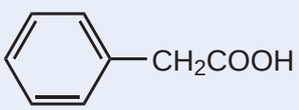
CARBOXYLIC ACIDS

Introduction

Carbon compounds containing a carboxyl function group, -COOH are called carboxylic acids. The Carboxyl group is the combination of carbonyl group $-\overset{\overset{O}{\|}}{C}-$ and the hydroxyl group (-OH).

However, carboxyl group has its own characteristic reaction. Carboxylic acids may be aliphatic (R – COOH) or aromatic (Ar – COOH) depending on the alkyl or aryl group attached to carboxylic carbon. Some higher members of aliphatic carboxylic acids (C_{12} to C_{18}) known as fatty acids occur in natural fats as esters of glycerol.

12.8 IUPAC nomenclature of Carboxylic acids

Compound (common name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
Formic acid HCOOH methanoicacid	–	meth	ane	oicacid
Acetic acid CH_3COOH Ethanoic acid	–	eth	ane	oicacid
Isobutyric acid $(CH_3)_2CHCOOH$ 2 – methylpropanoic acid	2 – methyl	prop	ane	oicacid
Phenyl acetic acid  2-phenyl ethanoic acid	2-phenyl	eth	ane	oicacid
Oxalic acid HOOC - COOH ethane–1, 2 – dioicacid	–	eth	ane	1, 2 – dioicacid

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